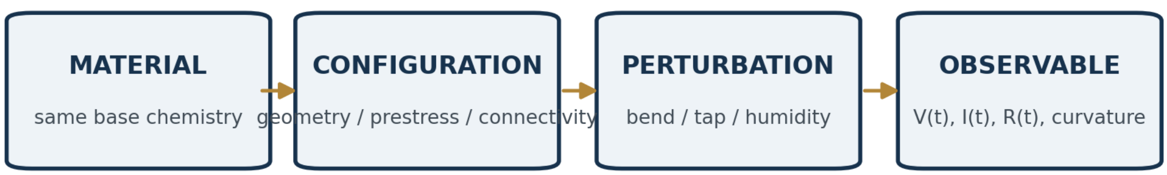
CONFIGURED-MATERIAL TRANSDUCTION

An Open Garage-Scale Replication Protocol

Chitosan-graphite films for geometry-, prestress-, and humidity-coupled electrical response



Observation is primary; mechanism models are tested against the reproducible map.

PURPOSE: A low-cost, inspectable experiment designed to establish a reproducible configuration-response effect without requiring agreement on one privileged microscopic explanation.

Protocol edition 1.0 | August 2026

How to use this document

This manual separates three things that scientific discussion often conflates: the observation, the reproducibility claim, and candidate descriptions of the mechanism. The experiment is deliberately designed so that a person may reject the proposed explanatory model and still reproduce, measure, report, or falsify the effect.

Read first	Use when
Quick-start protocol	You want to build the first coupons and run the experiment.
Measurement discipline	You want data that can survive skeptical inspection.
Posited mathematical/topological description	You want a model that can organize and predict the counterintuitive configuration dependence without being treated as mandatory ontology.
Appendix A - Bill of materials	You need sourcing, substitutions, and budget tiers.
Appendix B - Lab and skill requirements	You need workspace, safety, instrumentation, and competence thresholds.
Appendix C - Noise and artifact diagnostics	You need to understand thermal, humidity, shock, static, cable, contact, and instrument effects.
Appendix D - Replication report	You are publishing a successful or unsuccessful replication.

CLAIM BOUNDARY: The target result is not "geometry creates energy" and not "the signal must be piezoelectric." The target result is that intentionally changing the configured state of an otherwise matched material system produces a reproducibly distinguishable measured response under stated perturbations and environmental conditions.

Evidence status

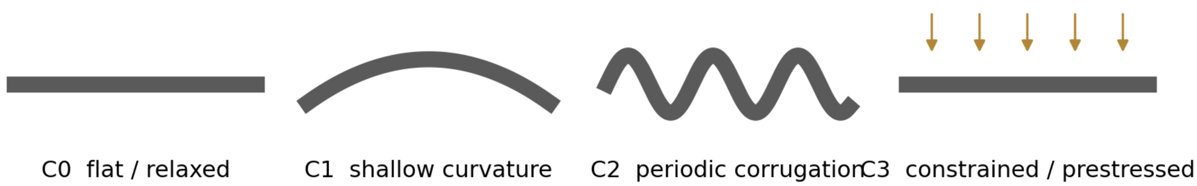
The accompanying research program treats chitin/chitosan as a spatially programmed multiphysics platform and explicitly distinguishes demonstrated primitives, transferred principles, derived consequences, and proposed architectures. It also requires system-level claims to remain traceable through composition, process history, geometry, parameter field, operator, state, energy source, measurement, uncertainty, and failure condition [S1].

Independent 2026 studies support two points directly relevant to this demonstrator: solution-cast chitosan can exhibit measurable piezoelectric response, with response strongly dependent on processing and crystallinity [1]; and hydration can strongly alter "piezoelectric-like" measurements, with the authors explicitly emphasizing other charge-generation mechanisms and measurement reliability [2]. The recent graphene-nanowrinkle literature provides a separate comparator showing that extreme curvature and strain gradients can generate pronounced flexoelectric polarization at the nanoscale [3]. That comparator motivates a geometry-sensitive hypothesis; it does not prove that a garage-scale chitosan-graphite wrinkle operates by the same microscopic mechanism.

1. The one-page experiment

The fastest defensible demonstration uses matched coupons from one master batch, a simple repeatable deflection jig, fixed electrodes, and environmental logging. Start with the mechanical test. Add humidity only after the baseline procedure works.

Matched coupon family



Best first test: same batch, same dimensions, same electrodes, controlled change in configuration.

1. Prepare 100 mL of approximately 1% acetic-acid solution. A practical household route is 20 mL of labeled 5% white vinegar plus 80 mL distilled water.
2. Dissolve 1.50 g chitosan into that solution. Stir until a uniform viscous solution forms. Several hours of stirring is acceptable; avoid boiling.
3. Add 0.15 g fine graphite powder (10 wt% relative to dry chitosan). Mix thoroughly immediately before casting. Record the graphite source and grade.
4. Cast one master sheet or several equal-volume films on clean glass or smooth plastic. Keep wet thickness, drying temperature, and drying time as consistent as practical.
5. Dry at room temperature for 24-48 h, protected from dust. If a low-temperature drying box is used, keep it below 50 °C for this baseline protocol and record the actual temperature.
6. Cut matched strips, nominally 20 mm x 60 mm. Reject visibly cracked coupons from the primary comparison but photograph and retain them as failure data.
7. Apply identical copper-tape or aluminum-foil electrodes. For the first protocol, place 10 mm-long electrodes at the two ends, leaving a central free span.
8. Condition all specimens together in the same room or closed container for at least 60 min before testing. Record temperature and relative humidity.
9. Place a coupon in the three-point deflection jig. Move the center contact between the same two stops for 20 load/unload cycles while recording voltage. Avoid touching electrodes or bare leads during acquisition.
10. Repeat for all configuration classes. Preserve raw waveforms, not only peak values. Report failures, drift, unexpected polarity changes, and null results.

RECOMMENDED FIRST SUCCESS CRITERION: A configuration-dependent effect is provisionally demonstrated when matched specimens or matched states show a repeatable difference in a predeclared response metric across repeated cycles, and the difference survives basic artifact checks. Independent reproduction upgrades the claim substantially.

2. Fabrication protocol

2.1 Why one master batch matters

Chitosan behavior depends on degree of deacetylation, molecular mass, water content, crystalline state, orientation, additives, thickness, and conditioning history [S1]. A household-grade experiment cannot eliminate those variables. It can, however, prevent them from silently differing between the primary comparison groups. The most important low-cost control is therefore a common master batch followed by matched processing.

2.2 Baseline formulation

Item	Baseline	Record
Chitosan	1.50 g	source, lot if available, stated degree of deacetylation/molecular weight if supplied
Acid solution	100 mL at ~1% acetic acid	vinegar acidity or reagent concentration
Graphite	0.15 g	source, powder type, nominal purity/particle information if available
Wet cast volume	same for every casting area	mL and casting area
Drying	24-48 h room temperature	temperature, RH, airflow, duration

This formulation is intentionally conservative and accessible. It is a starting point, not an optimized material. Once the baseline reproduces, graphite fraction, chitosan concentration, neutralization, annealing, electrode geometry, and hydration can become explicit experimental axes rather than hidden variations.

2.3 Mixing and casting

1. Use a clean glass jar with a lid. Add the acid solution first, then sprinkle chitosan gradually while stirring to reduce clumping.
2. Allow time for hydration and dissolution. If the mixture foams, let bubbles rise before casting. Do not interpret a partially dissolved, visibly heterogeneous batch as equivalent to a clear or uniformly translucent one.
3. Weigh graphite separately. Add it slowly while stirring. Fine carbon powder is messy; avoid airborne dust and clean spills wet rather than blowing them.
4. Mix immediately before each pour because graphite can redistribute or settle. If multiple plates are cast, alternate their order across intended configuration classes so that casting order does not become a hidden variable.
5. Use a taped perimeter, doctor-blade spacer, fixed-volume pipette/syringe, or marked spoon to keep deposited volume per area approximately constant.
6. Photograph the wet films and the dry films from the same distance with a scale reference.

2.4 Optional neutralization - not part of the first baseline

NaOH post-treatment is supported in the recent chitosan piezoelectric literature as a processing variable that can alter crystalline structure and measured response [1]. Because neutralization also changes chemistry, hydration, modulus, and ionic state, do not mix treated and untreated coupons in the first geometry-only comparison. Run it later as a separately labeled branch. Baking-soda treatment may be explored as a mild household alternative, but it should not be described as equivalent to a literature NaOH protocol without direct validation.

3. Program the configuration

The experiment has two complementary modes. Mode A changes geometry reversibly while using the same specimen. Mode B writes geometry or prestress during conditioning/drying. Mode A offers the strongest low-cost argument that composition alone does not determine the response; Mode B tests the more ambitious "field-programmed" material idea.

3.1 Mode A - paired within-coupon geometry

State	How to impose it	Why it matters
A0 flat	Clamp on a flat support with a fixed free span.	Reference state.
A1 shallow arc	Mount over a large-radius former, e.g. 75-100 mm radius.	Changes curvature with minimal sharp folding.
A2 tighter arc	Mount over a smaller-radius former, e.g. 30-50 mm radius, without cracking.	Tests stronger curvature.
A3 returned flat	Return the same coupon to the A0 fixture.	Tests reversibility and drift/history.

For the first paired experiment, use a small perturbation superimposed on each mounted state. The important comparison is not merely peak voltage but whether the response changes systematically with imposed configuration and returns toward baseline after the configuration is removed.

3.2 Mode B - written geometry/prestress

Class	Preparation	Interpretation boundary
C0 flat relaxed	Dry on a flat surface with minimal constraint.	Baseline processing state.
C1 curved	Dry or humidity-condition over a cylindrical former.	Geometry and process history both differ.
C2 corrugated	Dry against a reproducible corrugated former or constrain into periodic curvature.	Curvature field is spatially patterned.
C3 prestressed	Dry under gentle tension or constraint, then release.	Residual stress may differ even if final visible shape is similar.

IMPORTANT DISTINCTION: Curvature is geometry. A wrinkle does not automatically change mathematical topology. Topology enters when connectivity, boundaries, cracks, conductive contacts, or network loops change. This document keeps those concepts separate and then models their coupling.

4. Electrodes and measurement

4.1 Electrode geometry

Begin with end electrodes because they are easy to reproduce. Use the same metal, adhesive, area, and overlap on every coupon. Measure end-to-end resistance after electrode attachment. If resistance is beyond the meter range, record that fact rather than assuming a continuous graphite network. A later through-thickness configuration can use electrodes on opposite faces, but it constitutes a different experiment.

4.2 Three measurement tiers

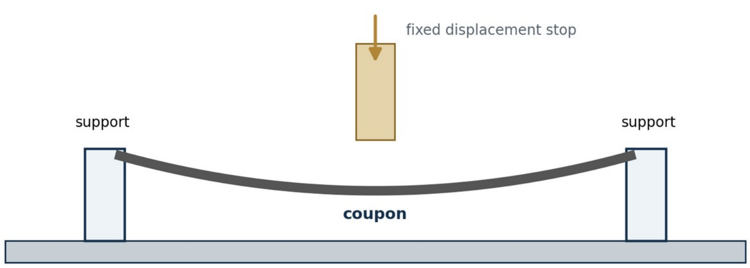
Tier	Instrumentation	What it establishes	Limit
1 - visual	Digital multimeter on sensitive DC voltage range	Immediate sign that deformation is associated with an electrical transient.	Slow update, input loading, no full waveform; use as demonstration only.
2 - recorded	High-input-impedance buffer + ADC/microcontroller or USB data logger	Time-resolved repeated cycles and environmental correlation.	Requires simple electronics and calibration.
3 - characterization	Electrometer/charge amplifier, oscilloscope/DAQ, force/displacement sensor	Quantified charge, impedance, bandwidth, phase, and artifact separation.	Higher cost and skill.

4.3 Recommended low-cost recorded front end

For an inexpensive reproducibility rig, use a CMOS-input voltage buffer before the ADC. A part such as the LMC6482 family has extremely low input-bias current and rail-to-rail operation, making it suitable for high-impedance sensing. A 16-bit ADS1115-class ADC can provide up to 860 samples/s and programmable gain; an inexpensive microcontroller can stream timestamped data. The ADC alone should not be assumed to equal an electrometer: the buffer, bias network, cable layout, input protection, and measurement range still matter.

Example currently available components and specifications are listed in Appendix A; these are sourcing suggestions, not required brands.

4.4 Mechanical tester



Simple three-point deflection jig: the goal is repeatable displacement, not expensive force metrology.

1. Use a rigid base with two rounded supports defining a constant span, nominally 40 mm for a 60 mm strip; use a rounded center contact and avoid sharp edges.
2. Provide hard displacement stops. Start with 2-5 mm center deflection and reduce it if the coupon cracks or permanently yields.
3. Operate the lever without touching the specimen or leads. Keep the motion rate approximately constant; one full load/unload cycle every 2 seconds is adequate for the baseline. A later load cell may be added without changing the geometry.

5. Mechanical replication experiment

5.1 Pre-test stabilization

- Place all coupons, fixture, and electrodes in the test room for at least 60 minutes.
- Record room temperature and RH at the beginning and end; continuous logging is preferred.
- Fix cables to the base so they cannot swing when the jig moves.
- Discharge obvious static from the operator/work surface before acquisition; do not touch sensing nodes during a run.
- Record 30-60 seconds of untouched baseline before the first cycle.

5.2 Run sequence

1. Randomize coupon order if several samples are tested.
2. Acquire 20 cycles for each specimen/state.
3. After each 20-cycle block, acquire 30 seconds of untouched baseline.
4. Repeat the entire sequence once after at least 10 minutes.
5. For Mode A, finish by returning the same coupon to its original flat state.
6. If practical, have a second person relabel samples so the operator does not know the configuration class during analysis.

5.3 Predeclare a simple response metric

Preserve the entire waveform, but choose one primary metric before inspecting group differences. Suitable low-cost choices include peak-to-peak voltage within a fixed time window, RMS voltage over the loading interval, or integrated current/charge if current is measured. A simple voltage metric is:

S_V = max[V(t)] - min[V(t)]   over the fixed cycle window

If current is recorded with a validated current/charge front end:

Q = integral from t0 to t1 of I(t) dt

For a configuration ratio:

R_G = median(S_configured) / median(S_flat)

The ratio is descriptive. Do not define "success" simply as R_G > 1. A suppressed or polarity-reversed response can be just as informative if it reproduces.

5.4 Minimum plots

- Raw V(t) for the full run, including untouched baseline.
- Overlay of several individual load/unload cycles without smoothing.
- Distribution of the predeclared metric for each configuration.
- Response versus imposed curvature state for the within-coupon Mode A test.
- Temperature and RH versus time whenever environmental logging is available.

6. Ambient-gradient extension

Only after the baseline mechanical experiment is understood should the environment replace the hand/jig as the changing input. Chitosan interacts strongly with water, and 2026 measurements show that hydration state can alter both mechanical behavior and apparent piezoelectric-like response [2]. The monograph likewise treats humidity as capable of changing mass, modulus, curvature, capacitance, ionic conductivity, and open-circuit voltage in asymmetric structures [S1].

6.1 Simple chamber

1. Use a clear food-storage container large enough to hold the mounted coupon without touching its walls.
2. Place the humidity/temperature sensor near the coupon but not in contact with it.
3. Route electrical leads through a small gap or sealed feedthrough and immobilize them.
4. Acquire a dry/stable baseline. Introduce a small open cup or damp pad of distilled water without touching the sample; close the chamber.
5. Record RH(t), T(t), and V(t) continuously as humidity rises.
6. Remove the moisture source or open the chamber and record the return trajectory.
7. Repeat with the same specimen geometry and with a different geometry.

DO NOT USE BREATH AS THE QUANTITATIVE PROTOCOL: Breathing on the sample is excellent for a visible demonstration but introduces heat, droplets, ions, airflow, motion, and electrostatic/contact variables at once. Use it only after the closed-chamber experiment has established the effect.

6.2 Asymmetric hydration architecture

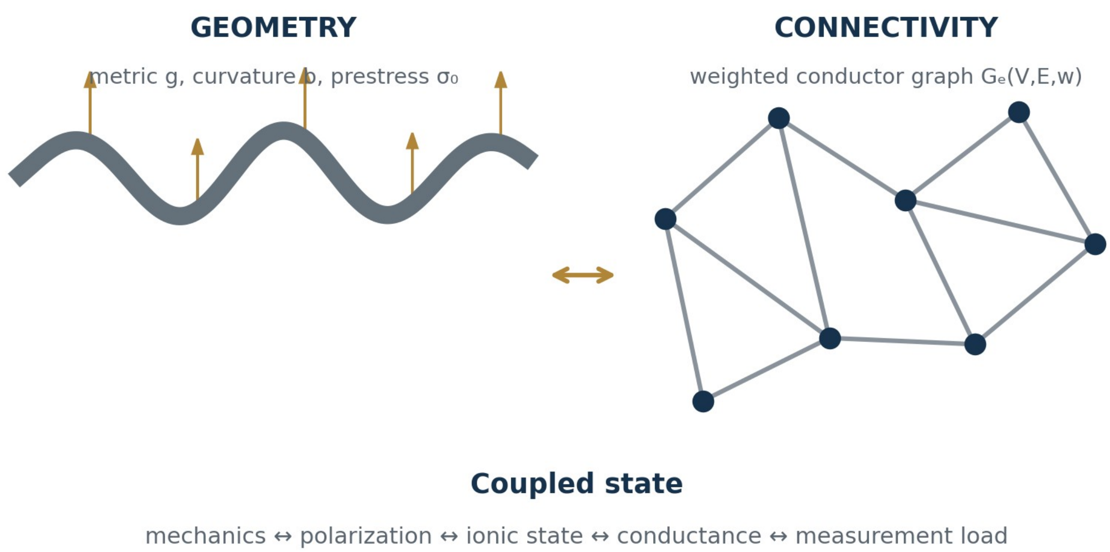
A second-stage specimen may expose one face to humidity while partially sealing the other, or may use a deliberately graded graphite/chitosan layer. The observational trajectory becomes:

$$\{ RH(t), T(t), \mathrm{kappa}(t), V(t), I(t), R(t) \}$$

This record does not require a single causal decomposition. It gives competing models a common dataset to explain.

7. Posited mathematical and topological description

STATUS OF THIS SECTION: This is a proposed descriptive framework that organizes the observations and makes predictions. It is not a condition for accepting the experimental result, and it is not presented as the unique microscopic explanation.



7.1 The empirical map comes first

Let p(x) denote the fabricated parameter field - composition, thickness, orientation, graphite fraction, electrode placement, prestress, and local transport properties. Let r(x,t) denote imposed mechanical, thermal, humidity, electrical, and chemical stimuli. Let s(x,t) denote internal state and B denote boundary conditions. Then the experiment can be written without choosing a privileged mechanism as:

$$y(t) = H[p(x), r(x,t), s(x,t), B]$$

The measured output y may contain displacement, curvature, voltage, current, resistance, temperature, and humidity. The central replication question is whether changing a controlled portion of p or B changes H reproducibly.

7.2 Geometry changes the mechanical operator

At small strain, a generic continuum description is:

$$\rho u_{tt} - \mathrm{div}(\sigma) = f$$

$$\sigma = C(x,z) : [\epsilon - \epsilon_0 - \epsilon_{th} - \epsilon_h] + \sigma_v$$

The terms epsilon_0, epsilon_th, and epsilon_h can represent residual/prestrain, thermal strain, and hydration-associated strain. After discretization, the vibrational/configurational problem takes the familiar form:

$$K(p,z)\,\phi_n = \omega_n^2\, M(p,z)\,\phi_n$$

Changing thickness, orientation, curvature, attachments, prestress, or boundary constraints changes K and sometimes M. Therefore two objects made from the same nominal chemistry need not have the same transfer function. This is the non-mysterious mathematical core of the geometry argument and follows the operator framing developed in the accompanying chitin work [S1,S2].

7.3 Curvature as a strain-gradient coordinate

For a thin sheet represented by a mid-surface, geometry can be described locally by a metric g_{ab} and curvature/second-fundamental-form b_{ab}. Mean curvature and Gaussian curvature are:

$$H = (1/2)\,\mathrm{tr}(g^{-1} b) \qquad K_G = \det(g^{-1} b)$$

In a thin-shell approximation, bending gives a through-thickness strain variation of the form:

$$\epsilon_{ab}(z) \approx \epsilon_{ab}^0 - z\, b_{ab}$$

so that a strain gradient scales with curvature:

$$\partial \epsilon_{ab} / \partial z \approx -b_{ab}$$

This relation explains why curvature is a natural coordinate for testing any mechanism sensitive to strain gradients. It does not by itself identify the electrical mechanism.

7.4 Candidate electromechanical expansion

A broad phenomenological polarization expansion may be written schematically as:

$$P_i = d_{ijk}\,\sigma_{jk} + \mu_{ijkl}\,(\partial \epsilon_{jk} / \partial x_l) + \chi_{ij} E_j + P_{i,\mathrm{ion}} + P_{i,\mathrm{other}}$$

The first term represents a piezoelectric-type contribution; the second a flexoelectric strain-gradient contribution; the third dielectric response; and the remaining terms leave room for ionic, interfacial, electrochemical, triboelectric, and other effects. The decomposition is deliberately non-exclusive. Recent chitosan work shows why such caution matters: hydration can profoundly change apparent response, and the authors explicitly use "piezoelectric-like" terminology where mechanisms are not fully separated [2].

7.5 Electrical collection and network topology

Graphite introduces an additional level: the electrical collection path may behave as a weighted network. Represent conductive regions or contacts as a graph $G_e=(V,E,w)$, where each edge has conductance w_e. The nodal relation is approximately:

$$L_G\, v = i$$

where L_G is the weighted graph Laplacian. Deformation can change edge weights through particle contact, separation, cracking, or compression. If an edge appears or disappears, the graph topology itself can change. If only its conductance changes, connectivity may remain the same while the weighted geometry of transport changes.

7.6 Geometry is not topology

A smooth wrinkle can greatly alter curvature while leaving the sheet topologically equivalent to a rectangle. Mathematical topology becomes directly relevant when the number of connected components, holes/loops, cracks, bridges, electrode-connected paths, or boundaries changes. For a conductor graph, simple descriptors include β_0 (number of connected components) and β_1 (independent loops). These need not be measured in the garage protocol, but they clarify what a future imaging/percolation model would mean by a genuine topological transition.

7.7 Coupled-state model

A compact state-space bookkeeping model consistent with the accompanying program is:

$$C_e(z,p)\,\dot{v} + G_e(z,p)\,v + i_{nl}(v,z,p) = B_e\,u_e + i_{em}(\dot{q}, \nabla\epsilon, z)$$

$$\dot{z} = g(z, q, \dot{q}, v, T, \mu_w; p)$$

$$y = h(q, \dot{q}, v, z, T, RH; p)$$

Here z can absorb hydration, ionic, contact, hysteretic, and other slow states. The model predicts a key experimental feature without forcing a microscopic ontology: if geometry or prestress changes the mechanical state q, strain distribution, electrical collection network, or internal state evolution, then the measured transfer function can change even when bulk composition remains nominally unchanged.

7.8 The prediction that matters

Define a controlled family of configurations C(lambda), where lambda might be curvature, corrugation amplitude, residual prestress, or boundary constraint. The model predicts an empirical response surface:

$$R(\lambda, P, E, \text{history}) = F[C(\lambda), P, E, \text{history}]$$

The strongest low-cost demonstration is therefore not one large pulse. It is a reproducible map in which changing lambda changes the response distribution, and returning lambda toward its original value returns the response toward its earlier state within the limits imposed by hysteresis and drift.

8. Commonly accepted explanations - and where uncertainty remains

The following mechanisms are not mutually exclusive. Their purpose here is to show how established physics can account for different portions of the measured behavior, while leaving the experiment free to reveal combinations, limits, or missing variables.

Candidate description	Why it is plausible	What would strengthen it	What could mimic it
Piezoelectric coupling	Chitosan films have measured electromechanical response; processing/crystallinity matter [1].	Polarity/strain symmetry, calibrated d-coefficient measurements, processing dependence.	Electrostatic, triboelectric, ionic, contact artifacts.
Flexoelectric / strain-gradient coupling	Polarization can couple to strain gradients; curvature naturally creates gradients. Graphene nanowrinkles show a strong nanoscale comparator [3].	Response scaling with curvature/strain gradient under controlled chemistry and contacts.	Piezoelectric bending, contact changes, geometry-dependent capacitance.
Hydration / ionic redistribution	Chitosan is hydrophilic; 2026 measurements show hydration strongly changes apparent response [2].	Controlled RH cycles, mass/curvature tracking, ion-blocking or electrode controls.	Thermal drift, condensation, leakage, electrochemistry.
Conductive-network modulation	Graphite contacts can form a deformation-sensitive network.	Four-wire resistance, loading dependence, microscopy/percolation study.	Electrode/contact resistance, cable movement.
Capacitive / dielectric modulation	Shape and separation change capacitance and electric field distribution.	Known bias/charge tests, capacitance measurement versus geometry.	Leakage, dielectric absorption.
Triboelectric / contact electrification	Contact/separation can generate large signals in polymers.	Noncontact deformation, material substitutions, grounding/shielding.	Can dominate a poorly controlled finger-tap experiment.
Electrochemical/interfacial potentials	Water, ions, graphite, and metal electrodes allow interfacial potentials.	Electrode-material swap, blocking electrodes, open-circuit aging, impedance spectroscopy.	Slow drift mistaken for mechanical generation.

8.1 Why the graphene work is relevant but not dispositive

Iyengar et al. report direct experimental and theoretical evidence of large intrinsic flexoelectricity in graphene nanowrinkles with sub-nanometer curvature, including curvature-dependent work-function shifts and reproducible currents [3]. The relevance here is conceptual: extreme geometry can change electromechanical response through strain-gradient physics. The scale, material, electronic structure, and boundary conditions are radically different from a millimeter-scale chitosan-graphite film. Therefore the graphene result supports the plausibility of a geometry-sensitive description but cannot be used as proof of the garage demonstrator's mechanism.

8.2 Why hydration deserves equal attention

Akbarinejad et al. report that fully dried chitosan films showed no d33 signal in their measurement while fully hydrated films showed a much larger response, and they emphasize the challenge of other charge-generation mechanisms [2]. That result makes hydration both a candidate mechanism and a major confound. In this protocol, humidity is therefore measured even during the "mechanical" experiment rather than treated as background.

9. Noise, artifacts, and environmental sensitivity

Small electrical signals are extraordinarily easy to manufacture accidentally. The goal is not to eliminate every disturbance before beginning; it is to know which disturbances can impersonate the target effect and to record enough context that they can be diagnosed.

Disturbance	Typical symptom	Low-cost mitigation	Diagnostic test
Temperature drift	slow baseline wander; changing resistance/offset	allow 60 min equilibration; log T; avoid lamps/heaters	repeat with no mechanical motion while T changes
Humidity drift	slow voltage/resistance/curvature changes	log RH continuously; condition all samples together	hold fixture still through an RH cycle
Mechanical shock / table vibration	spikes unrelated to commanded cycle	heavy base, foam isolation, fixed jig	tap table away from sample and record signature
Cable motion / microphonics	spikes when wires move	short twisted leads; strain relief; tape cable to base	move cable with specimen untouched
Triboelectric contact	large polarity-dependent spikes on touch/separation	do not touch during run; rounded fixed contact; material control	operate fixture without sample or with inert plastic
Electrostatic pickup	large response to nearby hands/mains	shielded enclosure, battery power, grounded shield	approach hand without moving sample
Electromagnetic pickup	50/60 Hz periodic signal	short leads, twist pair, shielding, battery supply	FFT or observe periodicity with fixture untouched
Electrode/contact resistance	step changes, hysteresis, erratic polarity	consistent tape pressure/area; inspect before/after	measure resistance before and after run
Instrument loading	signal collapses on one meter but not another	high-input-impedance buffer; state input resistance	compare meters/buffer; add known resistor
ADC clipping / aliasing	flat-topped or distorted peaks	set range/gain; adequate sample rate	inspect raw counts and repeat at lower gain
Graphite settling / batch gradient	systematic differences by casting order	stir before each cast; randomize class order	compare coupons by pour sequence
Film cracking / yielding	irreversible response jump	lower displacement; inspect optically	return-flat test; resistance before/after

A USEFUL NEGATIVE CONTROL: Run the complete mechanical cycle with the fixture empty, then with an inert plastic strip carrying the same electrode wires. The purpose is not to prove a mechanism; it is to learn the electrical signature of the apparatus itself.

10. Evidence ladder and falsification

Level	Result	What can be said
0	Meter moves once.	Anecdote only.
1	Repeated signal synchronized with deformation on one specimen.	A reproducible specimen-level association exists.
2	Matched coupons/states show separated response distributions.	Configuration dependence is supported within the batch.
3	Return-to-state and artifact controls preserve the difference.	Simple handling/instrument artifacts become less plausible.
4	Blindly labeled configurations can be classified above chance.	The response contains reproducible information about configuration.
5	Independent builders reproduce the relationship from the public protocol.	The effect is portable beyond the originating setup.
6	Predictive response surface survives new geometries/environmental trajectories.	The proposed descriptive model gains predictive power.

10.1 What would count against the effect?

- Configured and control samples remain statistically indistinguishable across repeated batches.
- The apparent signal disappears when cable movement, touching, or instrument loading is controlled.
- The response follows casting order, electrode batch, or operator rather than intended configuration.
- Returning the same coupon to its original configuration does not restore the earlier response and the difference is fully explained by damage or irreversible drying.
- Independent replications using adequately matched materials and procedures consistently fail to recover the reported relationship.

A failed replication is not an embarrassment to this program. It narrows the conditions under which the effect exists and improves the eventual model.

11. Open replication invitation

This protocol is intended to move the question out of authority contests and into inspectable practice. Institutions, independent laboratories, classrooms, makerspaces, and garage experimenters can contribute equally useful evidence if they disclose methods and raw observations.

PLEASE REPORT NEGATIVE RESULTS: If you follow the protocol and cannot reproduce the effect, publish that result with the same care as a positive result. State what materials were used, what differed, the environmental conditions, the instrument, and the raw traces.

Questions for every replicator

- Did you obtain a time-locked electrical response to the mechanical cycle?
- Did the same coupon produce a distinguishable response when its imposed geometry changed?
- Did the response return when the geometry returned?
- Did written/prestressed geometries differ from flat controls?
- How strongly did temperature and humidity affect the result?
- Did resistance change with bending or hydration?
- Which artifact checks changed the result?
- Which candidate explanation best predicted your observations?
- What observation did the proposed model fail to anticipate?
- Could a blinded analyst identify configuration from the recorded response?

The common object is the reproducible experiment. A model earns standing by compressing the observations, predicting new ones, and surviving attempts to break it.

Appendix A - Bill of materials and sourcing

The protocol is designed around ordinary materials. Exact brands are not mandatory; record substitutions. Prices below are approximate US retail snapshots or broad budget ranges in August 2026 and should not be treated as quotes.

Item	Minimum specification / substitution	Where to look	Approx. budget
Chitosan powder	Powder, preferably with stated degree of deacetylation and molecular weight; food/supplement grade is acceptable for entry-level replication if fully documented.	Laboratory chemical supplier; bulk supplement supplier. A current consumer example is BulkSupplements chitosan powder.	$10-35
White vinegar / acetic acid	Labeled 5% distilled white vinegar for baseline; reagent acetic acid optional.	Grocery store; lab supplier.	$2-8
Distilled water	Commercial distilled/deionized water.	Grocery/pharmacy.	$1-3
Fine graphite	Fine powdered graphite; avoid charcoal. Artist-grade powder is an accessible option.	Art supplier; industrial lubricant supplier. Blick lists artist-quality graphite powder.	$8-25
Scale	0.01 g resolution preferred.	Kitchen/online electronics retailer.	$10-20
Casting plates	Glass picture-frame sheet, microscope glass, smooth PET/acrylic.	Hardware/craft store.	$0-10
Copper tape / foil	Conductive copper tape or aluminum foil; use identical material within a run.	Electronics/craft/hardware.	$5-12
Digital multimeter	10 MOhm-class voltage input or better preferred; mV range.	Hardware/electronics retailer.	$15-40
Humidity/temperature sensor	AHT20/SHTC3/SHT31-class digital sensor.	Electronics maker supplier. Adafruit currently lists AHT20-class boards around the low-single-digit dollar range.	$5-15
Microcontroller	USB-capable Arduino/RP2040-class board.	Maker/electronics supplier.	$5-15
16-bit ADC	ADS1115-class breakout, programmable gain, up to 860 samples/s.	Maker/electronics supplier; Adafruit currently lists an ADS1115 breakout.	$10-20
High-Z buffer op amp	CMOS-input dual op amp such as LMC6482-family DIP part or current equivalent.	DigiKey/Mouser/electronics distributor.	$2-10
Breadboard + resistors	Solderless board, 100 kOhm-10 MOhm resistors, capacitors, jumpers.	Electronics kit supplier.	$8-20
Jig materials	Wood/acrylic/cardboard, screws, rounded dowel/bolt, clamps.	Scrap/hardware store.	$0-15
Closed humidity chamber	Clear food-storage container.	Grocery/household.	$3-10
PPE	Safety glasses, nitrile gloves, dust mask for powder handling as appropriate.	Hardware/pharmacy.	$5-15

A.1 Practical budget tiers

Tier	What you buy	Expected total
Basic visual	Consumables + scale + copper tape + DMM + improvised jig	$35-80 if you already own common tools
Recorded	Basic tier + microcontroller + RH/T sensor + ADC + buffer electronics	$70-140
Characterization	Recorded tier + better DAQ/electrometer/oscilloscope/load cell	Open-ended; not required for first replication

Appendix B - Lab, safety, and skill requirements

Requirement	Entry level	Recorded replication	Advanced
Workspace	Clean table, ventilation, washable surface	Stable table away from fans/heaters	Environmental chamber / vibration-isolated bench optional
Chemical handling	Household vinegar, water, graphite, gloves/glasses	Same + careful weighing and labeling	NaOH post-treatment only with appropriate chemical handling
Mechanical fabrication	Cut film, tape electrodes, build simple lever jig	Repeatable stops, cable strain relief	Load cell, displacement sensor, machined fixture
Electronics	Use a DMM safely	Breadboard op amp/ADC, USB logging	Charge amplifier/electrometer/impedance spectroscopy
Data skill	Save readings and photos	CSV logging, plotting, median/spread, randomized order	Uncertainty propagation, regression, blinded classification
Minimum reasoning level	Able to follow a controlled procedure and change one variable at a time	Able to distinguish signal, artifact, and uncontrolled covariate	Able to build and test competing mechanistic models

B.1 Safety

- Wear eye protection during mixing, cutting, and any NaOH work.
- Avoid inhaling graphite dust; handle powder gently and clean with damp methods.
- Do not heat household containers on open flame. Use only low-temperature warming with appropriate vessels.
- If using NaOH, use a deliberately prepared dilute solution, gloves, eye protection, labeled containers, and appropriate disposal practices. NaOH is corrosive even though the baseline vinegar formulation is mild.
- Keep liquids away from powered electronics. Battery-powered measurement is preferred near the wet/humidity setup.
- Do not place sealed containers under pressure or heat.

B.2 Skill assessment

An entry-level but capable replicator should be able to weigh to 0.01 g, measure liquid volume, keep samples labeled, avoid cross-contamination, use a multimeter, follow a repeated motion protocol, save raw data, and resist changing several variables after seeing an interesting result. No advanced degree is required. Careful procedure is required.

Appendix C - Noise and measurement checklist

Before casting

- Record chitosan/graphite source and lot information available.
- Calibrate/check scale with a known mass if possible.
- Label every casting plate before pouring.

Before testing

- Photograph coupon with ruler.
- Record dimensions, electrode area, resistance, T, RH.
- Fix cables mechanically.
- Acquire untouched baseline.
- Verify ADC range is not clipping.

During testing

- Do not touch electrodes or leads.
- Use the same displacement stops.
- Keep cycle timing consistent.
- Write down shocks, door slams, fan changes, or handling mistakes.
- Do not delete bad cycles; flag them.

After testing

- Measure resistance again.
- Photograph cracks or permanent deformation.
- Export raw CSV before filtering.
- Retain configuration labels and randomization key.
- Publish negative/null data with positive data.

C.1 Suggested file structure

project / batch_YYYYMMDD / coupon_ID / raw / processed / photos / notes

Recommended CSV columns: timestamp_s, coupon_id, configuration, cycle_id, displacement_state, voltage_V, current_A_if_available, resistance_ohm_if_sampled, temperature_C, RH_percent, operator_event.

Appendix D - Replication report form

Replicator / lab / alias	
Date and location (city/country only)	
Chitosan source / grade / lot	
Graphite source / grade	
Acid concentration and preparation	
Chitosan concentration	
Graphite fraction	
Casting area and volume	
Drying temperature / RH / duration	
Post-treatment	
Coupon dimensions	
Electrode material / area / spacing	
Pre-test conditioning	
Fixture span and displacement	
Instrument and input impedance	
Sample rate / range	
Temperature / RH during run	
Primary predeclared response metric	
Number of coupons / cycles	
Artifact controls performed	
Result: reproduced / partial / not reproduced	
Most important unexpected observation	
Preferred explanatory model, if any	
What would you test next?	

OPEN INVITATION: Please report successful, partial, and unsuccessful replications. The strongest contribution may be the experiment that identifies a hidden dependency and makes the next version harder to fool.

References and source basis

[S1] Semita, L. Field-Programmed Chitin Metamaterials: Transduction Architectures, Reconfigurable Energy Routing, Rudimentary Material Logic, and Hybrid-Living Systems. Research Program Edition 1.0, August 2026. User-supplied monograph.

[S2] Semita, L. Geometrically Governed Chitin Interfaces: Resonance, Aerodynamic Coupling, Acoustic Transduction, and Collective Thermal Regulation in Beetles, Cicadas, and Honeybees. Working paper, July 2026. User-supplied manuscript.

[S3] Semita, L. The Pistol Shrimp Claw: A Biologically Encoded, Multi-Scale Energy-Routing System. Technical synthesis and engineering interpretation, July 2026. User-supplied manuscript.

[1] Papa, S.; Montorsi, M.; Arrighetti, L.; et al. "A Comprehensive Study of Piezoelectricity in Chitosan Films." Small Science 6(3), 2026, e202500593. DOI: 10.1002/smsc.202500593.

[2] Akbarinejad, A.; Fiedler, H.; Gito, D. A.; et al. "The role of water molecules in piezoelectric-like effects in chitosan-based biodegradable films." Journal of Materials Chemistry A 14, 21270-21281 (2026). DOI: 10.1039/D6TA01041E.

[3] Iyengar, S. A.; McHugh, J. G.; Salvage, J. P.; et al. "Sub-nm Curvature Unlocks Quantum Flexoelectricity in Graphene." arXiv:2503.21996v2 (2025). DOI: 10.48550/arXiv.2503.21996.

[4] Adafruit Industries. ADS1115 16-Bit ADC - 4 Channel with Programmable Gain Amplifier. Product documentation consulted August 2026.

[5] Adafruit Industries. AHT20/SHT-series temperature and humidity sensor documentation. Consulted August 2026.

[6] Texas Instruments / DigiKey. LMC6482 CMOS dual operational amplifier product specifications. Consulted August 2026.

Source-to-protocol traceability

Protocol element	Basis
Geometry as part of operator; prestress/boundary conditions alter response	[S1], [S2]
Broad coupled state description and humidity/ionic coupling	[S1]
Energy-routing language and multiscale configuration perspective	[S3]
Chitosan piezoelectricity and processing/crystallinity sensitivity	[1]
Hydration dependence and caution about piezoelectric-like measurements	[2]
Strong curvature/strain-gradient flexoelectric comparator	[3]
Low-cost logging component examples	[4]-[6]

BUILD IT • MEASURE IT • TRY TO BREAK IT • REPORT WHAT HAPPENED